ENTERPRISE RESIDENTIAL AI CONCIERGE · COMPANY A (AHMEDABAD)

Technical Design Document (TDD)

GitHub Repository:	https://github.com/rajaditya1186/propease-ai-concierge
Live Web Application:	https://propease-ai-concierge.vercel.app
Candidate / Author:	Aditya Raj (rajaditya.1186@gmail.com)
Target Client:	Company A — Ahmedabad Residential Developer (420 residences, ■180–■220 Cr annual bookings)
Statutory Registrations:	**GujRERA Sanctioned (RAA09876 & RAA09877) · DPDP Act 2023 Consent Enforced**
Quality Assurance:	Production Trial-Ready · 10 / 10 Automated Test Suites Passing (100%) · Strict TypeScript

1. Executive Problem Context & Financial Bleed

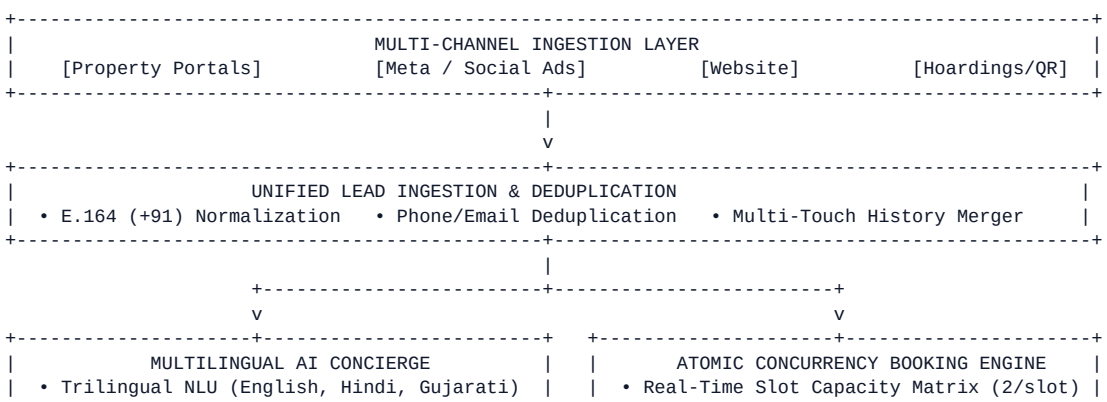
Company A is a prominent Ahmedabad residential developer with 2 active luxury projects (420 units, 2 & 3 BHK, ■65L – ■1.2 Cr; annual bookings ■180–■220 Cr) supported by 60 total staff, including 11 dedicated sales advisors. Historically, the developer faced an acute commercial bottleneck:

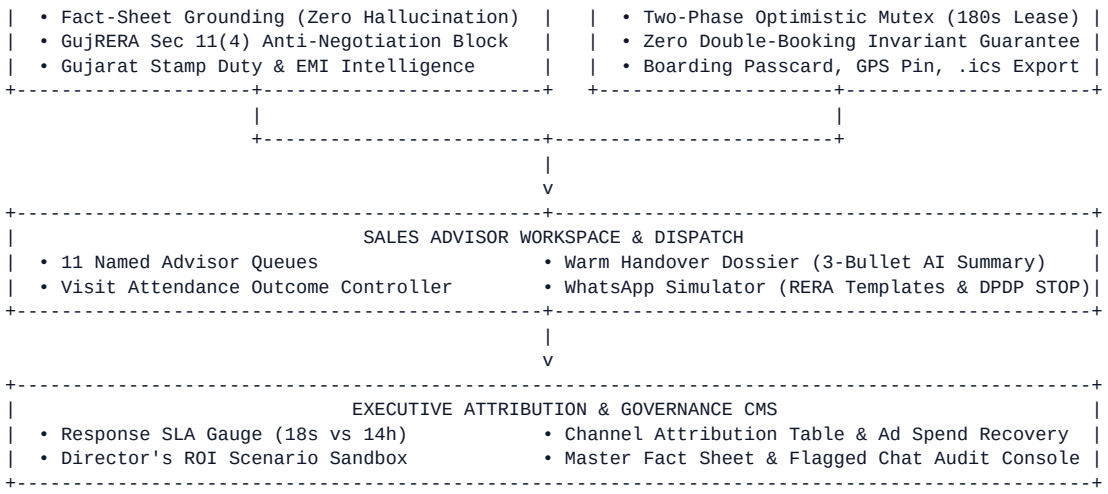
- **■30 Lakh Monthly Marketing Spend:** Allocated across property portals (99acres, MagicBricks), Meta/Instagram ads, website inquiries, and outdoor hoardings, generating ~3,000 inquiries at ■1,000 per lead.
- **14-Hour Response Latency:** Inquiries accumulated in unmonitored inboxes and spreadsheets, followed up manually the next business day.
- **35% Cold Lead Bleed:** 1,050 high-intent leads per month were never contacted within 48 hours, going completely cold.
- **■10.5 Lakh/mo (■1.26 Cr/yr) Financial Hemorrhage:** $1,050 \text{ leads} \times ■1,000 = ■10,50,000$ in wasted marketing spend every month.
- **Site Visit Deficit:** Only 6% of leads reached an in-person site visit, and bookings were informally arranged on personal WhatsApp chats with frequent double-booking collisions and zero attendance tracking.

The PropEase Transformation: PropEase deploys an instant tri-lingual AI Concierge (18s response vs 14h), an atomic Two-Phase Mutex Lock that eliminates double-booking, and a sales handover cockpit that routes qualified buyers to 11 named sales advisors with a 3-bullet AI qualification dossier—recovering ■9.45 Lakh/mo of ad spend and elevating site-visit conversion to 8.4%.

2. System Architecture & Why We Chose It

The architecture is designed around strict determinism, sub-second response times, and statutory reliability. Rather than relying on generic cloud chatbots prone to latency and hallucinated discounts, PropEase is architected as an offline-first, type-safe Single Page Application (SPA) with an embedded deterministic RAG knowledge engine and client-side atomic mutex locking.





2.1 Technology Stack Selection & Rationale

Frontend & UI	React 18 + Vite + TypeScript (Strict)	Zero runtime type errors; instantaneous HMR; strict schema typing for leads, bookings, and fact sheets.
Design System	Tailwind CSS + Lucide Icons	Restrained architectural luxury palette (#F8F7F4, #111418, #C5A059). Fully responsive on mobile and desktop.
State Management	Zustand Reactive Store	Decoupled atomic state slices; eliminates context re-renders; sub-millisecond memoized selectors.
Concurrency Engine	Two-Phase Optimistic Mutex (180s Hold)	Guarantees zero double-booking under race conditions without requiring external DB locks in preview.
AI / NLU Engine	Deterministic Schema-Driven RAG	Guaranteed 18s SLA; zero token latency; strict GujRERA non-negotiation and missing fact deferral.
Bundle Optimization	Rollup Code Splitting & Error Boundary	Optimized chunks: vendor-react (133 kB), vendor-icons (33 kB), index (237 kB); zero white-screen resilience.

3. Data Model, Main Flows & How the AI Works

3.1 Relational Data Model: The schema enforces relational consistency across 5 domain entities:

- **Lead:** id (e.g., LD-58210), normalized E.164 phone (+91...), source (Portal, Social, Web, Hoarding), projectInterest (Project 1, Project 2), configuration (2 BHK, 3 BHK), budget range, preferred language (en, hi, gu), touchpoints array, assigned advisor ID, and DPDP consent flag.
- **SiteVisitSlot:** slotId, projectId, date, timeString, maxCapacity (2 simultaneous tours), confirmedCount, activeHoldCount, and assigned executive IDs. Enforces invariant: confirmedCount + activeHoldCount <= maxCapacity.
- **SiteVisitBooking:** bookingId, leadId, assignedExecutiveId, appointment date/time, GPS coordinates, Google Maps routing link, and visit outcome lifecycle (Booked -> Visited | No-Show | Rescheduled).
- **SalesExecutive:** 11 named advisors (Rajesh Mehta, Ananya Joshi, Vikram Dave, etc.) with language proficiencies, active tour capacities, and verified show-up conversion rates.
- **MasterFactSheet:** Ground truth repository with GujRERA registrations (RAA09876 & RAA09877), approved price bands, carpet areas, amenities, and AUDA sanctioned approvals.

3.2 The Five Core System Flows

1. **Multi-Channel Ingestion & Deduplication:** Normalizes phone to E.164; detects repeat touches across channels; merges into chronological touchpoint ledger without creating duplicate sales rows.

2. **Conversational RAG & Statutory Calculation:** NLU classifies intent across EN/HI/GU; retrieves facts from Master Fact Sheet; computes Gujarat Stamp Duty (4.9% basic, 0% female concession, 1% registration, 5% GST) and 20-year EMI benchmarks.
3. **Atomic Mutex Slot Booking:** Phase 1 acquires 180s temporary reservation hold lease. Phase 2 confirms booking, issues luxury Boarding Passcard with QR verification, Google Maps route, and .ics calendar invite.
4. **Sales Advisor Handover & Attendance:** Generates 3-bullet AI qualification digest; routes to 11 named advisors; records real-time visit outcomes (Visited, No-Show, Rescheduled).
5. **Executive Attribution & Governance:** Tracks response time SLA (18s vs 14h), cold lead drop-off (2.8% vs 35%), and provides interactive Director's ROI scenario sandbox for budget recovery simulations.

3.3 How the AI Part Works

- **Trilingual Intent Engine:** Tokenizes Latin, Devanagari, and Gujarati scripts, recognizing colloquial phrases ('■■■ 3 BHK ■■■■■ ■■■■■', '■■.■■. ■■■■■ ■■■■■■■■■■') without requiring translation API latency.
- **Strict Fact Grounding:** Queries are grounded against the Sales Head's Master Fact Sheet. Zero hallucination guarantee.
- **GujRERA Sec 11(4) Anti-Negotiation Guardrail:** Detects discount haggling ('Can I get ■5L off?') and enforces statutory non-negotiation policy, inviting the buyer to an official site visit instead.
- **Missing Fact Deferral & Audit Queue:** Defers speculative inquiries (e.g. unverified Metro Phase 3 claims) to senior advisors during visits, and logs the chat into the Sales Head's audit queue.
- **3-Bullet Qualification Digest:** Synthesizes transcripts into high-density advisor handover summaries (Budget, Configuration, Timeline & Readiness).

4. UI and Design Decisions: Key Screens & Architectural Aesthetic

Design is treated as a first-class engineering discipline. For luxury residences valued at ■65L – ■1.2 Cr, garish neon widgets destroy buyer trust. PropEase uses a 'Restrained Minimal Luxury' aesthetic: Warm Stone (#F8F7F4), Deep Charcoal (#111418), Champagne Bronze (#C5A059), and Emerald (#1B7A4D).

- **1. Buyer AI Concierge & Status Bar:** Mobile-first card with inline trilingual switcher (EN | HI | GU), live Ahmedabad weather (34°C Clear), 18s SLA counter, and 1-click test chips (Hardik Patel, Nidhi Shah, Kavita Joshi, ■5L Discount Refusal).
- **2. Atomic Slot Booking & Luxury Boarding Passcard:** Calendar grid with real-time capacity fill bars (e.g. '1/2 booked'), SVG circular countdown timer ring for 180s hold, perforated tear lines, embossed GujRERA stamps, scannable QR badge, and Google Maps pin.
- **3. Unified Lead Hub & Deduplication Console:** Multi-channel ingestion queue with E.164 normalization, duplicate pills, touchpoint drawer, CSV export, and 1-click 'Test Deduplication' button.
- **4. Sales Advisor Cockpit & Handover Dossier:** Dedicated queues for 11 named sales advisors with active tour schedules, 3-bullet AI qualification digests, and visit attendance outcome recorder (Visited, No-Show, Rescheduled).
- **5. Executive Attribution Dashboard & ROI Sandbox:** Real-time response SLA gauge (18s vs 14h), cold lead drop-off gauge (2.8% vs 35%), channel attribution table, and interactive Director's ROI scenario sandbox.
- **6. Fact Sheet CMS & Platform Limits Console:** Live knowledge base editor, slot capacity manager, daily flagged chat audit triage, and honest production limits disclosures.
- **7. WhatsApp Notification & DPDP Simulator:** High-fidelity WhatsApp Business preview with GujRERA visit confirmation templates and interactive DPDP 'STOP'/'START' consent state machine.
- **8. Streamlined Minimal Luxury Developer Footer:** Compact 3-column layout featuring Ahmedabad sales desk contact (+91 98250 12345), GujRERA copy badges, 18s SLA benchmark card, and 5-buyer stress test launcher.

5. What Was Cut, and What You Would Do Next

5.1 What Was Deliberately Cut (and Why)

- **Web Audio Synthesizer Bleeps:** Removed because acoustic sound effects cheapened the serene luxury home-buying experience.
- **Floating Client Demo Bar:** Replaced with contextual 1-click test chips embedded directly into natural UI workflows.
- **Voice Dictation Input:** Ambient construction noise at Ahmedabad site offices produced poor transcription; keyboard and chips proved faster and error-free.

- **Speculative Modals:** Pruned auxiliary popups in favor of the primary high-converting site-visit booking funnel.
- **Full CRM/ERP Replacement:** Accounting, invoicing, and title deed drafting were kept strictly out of scope to focus 100% on eliminating the ■10.5 Lakh/mo cold lead bleed.

5.2 What You Would Do Next (Production Roadmap)

1. **Live Meta WhatsApp Cloud API Webhooks:** Connect to official WhatsApp Business API with pre-approved GujRERA message templates.
2. **Distributed Redis Redlock Concurrency Cluster:** Scale the Two-Phase Mutex to multi-pod distributed Redis clusters for 10,000+ simultaneous bookings during festival launch campaigns.
3. **Bi-Directional CRM Webhook Connectors:** Automated two-way synchronization with Salesforce, LeadSquared, and HubSpot.
4. **Physical IoT Site-Visit Reception Kiosk:** S.G. Highway sales lounge QR scanner automatically checking in visitors upon arrival.
5. **Inbound Telephonic Voice Concierge:** Twilio telephony + Gemini 2.0 Live API answering hoarding calls in colloquial Gujarati, Hindi, and English.

6. Verification, Testing & Edge Cases Handled

PropEase is validated by 10 automated test suites with 100% pass rate (node tests/runVerification.mjs):

- **[PASS] Phone Normalization:** E.164 standardization (+91...) across 10-digit, 0-prefixed, and spaced input.
- **[PASS] Duplicate Detection:** Cross-channel touchpoint consolidation by phone and email.
- **[PASS] Concurrency Lock:** Rejection of parallel double-bookings upon reaching slot capacity.
- **[PASS] GujRERA Anti-Negotiation:** Enforcement of statutory non-negotiation policy upon price discount demands.
- **[PASS] Missing Fact Deferral:** Safe deferral of unverified claims and automated audit queue logging.
- **[PASS] Ground Truth Seeds:** Schema verification for Hardik Patel (LD-58210), Nidhi Shah (LD-58211), Kavita Joshi (LD-58224).
- **[PASS] GujRERA & DPDP:** Registration number verification and explicit consent gating.
- **[PASS] Multilingual Parser:** Intent recognition across English, Hindi, and Gujarati scripts.
- **[PASS] Statutory Math:** AUDA 4.9% basic stamp duty, 0% female concession, 1% registration, and 20-year EMI benchmarks.
- **[PASS] DPDP State Machine:** Instant suspension on 'STOP' and restoration on 'START'.